SIMULATION EXPERIMENT – 2

Numerical Simulation and Performance Analysis of Microwave Transmission Lines and Two-Port Networks Using Scattering Parameters

AIM: To simulate a 50 Ω microwave two-port transmission line over 8–12 GHz, obtain \(S{11}, S{12}, S{21}, S{22}\), and calculate return loss, insertion loss, and transmission efficiency.The manual specifies a 50 mm line, a 10 GHz center frequency, and 50 Ω port terminations.

MATLAB program

```
clc;

clear;

close all;


% ECA13 - Experiment 2

% Microwave Transmission Line and Two-Port S-Parameters


c = 3e8;              % Speed of light (m/s)

Z0 = 50;              % Characteristic impedance (ohms)

L = 50e-3;            % Transmission line length (m)

f0 = 10e9;            % Center frequency (Hz)


% Frequency sweep: 8 GHz to 12 GHz in 10 MHz steps

f = (8e9:10e6:12e9);

N = length(f);


% Initialize S-parameters

S11 = zeros(1,N);

S22 = zeros(1,N);
```

```
S21 = zeros(1,N);
S12 = zeros(1,N);

% Ideal matched, lossless transmission line model
beta = 2*pi*f/c;
S21 = exp(-1j*beta*L);
S12 = exp(-1j*beta*L);

% Convert S-parameters to dB for plotting
S11_dB = 20*log10(max(abs(S11),1e-6));
S22_dB = 20*log10(max(abs(S22),1e-6));
S21_dB = 20*log10(abs(S21));
S12_dB = 20*log10(abs(S12));

% Find index corresponding to 10 GHz
[~,idx] = min(abs(f-f0));

% Calculate performance at 10 GHz
RL = -20*log10(max(abs(S11(idx)),1e-12));
IL = -20*log10(abs(S21(idx)));
efficiency = abs(S21(idx))^2*100;

fprintf('Center Frequency = %.2f GHz\n',f(idx)/1e9);
fprintf('Characteristic Impedance = %.2f ohms\n',Z0);
fprintf('Line Length = %.2f mm\n',L*1000);
fprintf('S11 = %.4f\n',abs(S11(idx)));
fprintf('S21 = %.4f\n',abs(S21(idx)));
fprintf('S12 = %.4f\n',abs(S12(idx)));
fprintf('S22 = %.4f\n',abs(S22(idx)));
```

```
fprintf('Return Loss = %.2f dB (ideal matched)\n',RL);
fprintf('Insertion Loss = %.2f dB\n',IL);
fprintf('Transmission Efficiency = %.2f %%\n',efficiency);

% Plot S-parameter magnitudes
figure;
plot(f/1e9,S11_dB,'r','LineWidth',2);
hold on;
plot(f/1e9,S21_dB,'b','LineWidth',2);
plot(f/1e9,S12_dB,'g--','LineWidth',2);
plot(f/1e9,S22_dB,'m--','LineWidth',2);
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('S-Parameter Response of a 50 Ohm Two-Port Network');
legend('S11','S21','S12','S22','Location','best');
grid on;
ylim([-130 5]);
hold off;

% Plot transmission phase
figure;
plot(f/1e9,unwrap(angle(S21))*180/pi,'LineWidth',2);
xlabel('Frequency (GHz)');
ylabel('S21 Phase (degrees)');
title('Transmission Phase versus Frequency');
grid on;
```

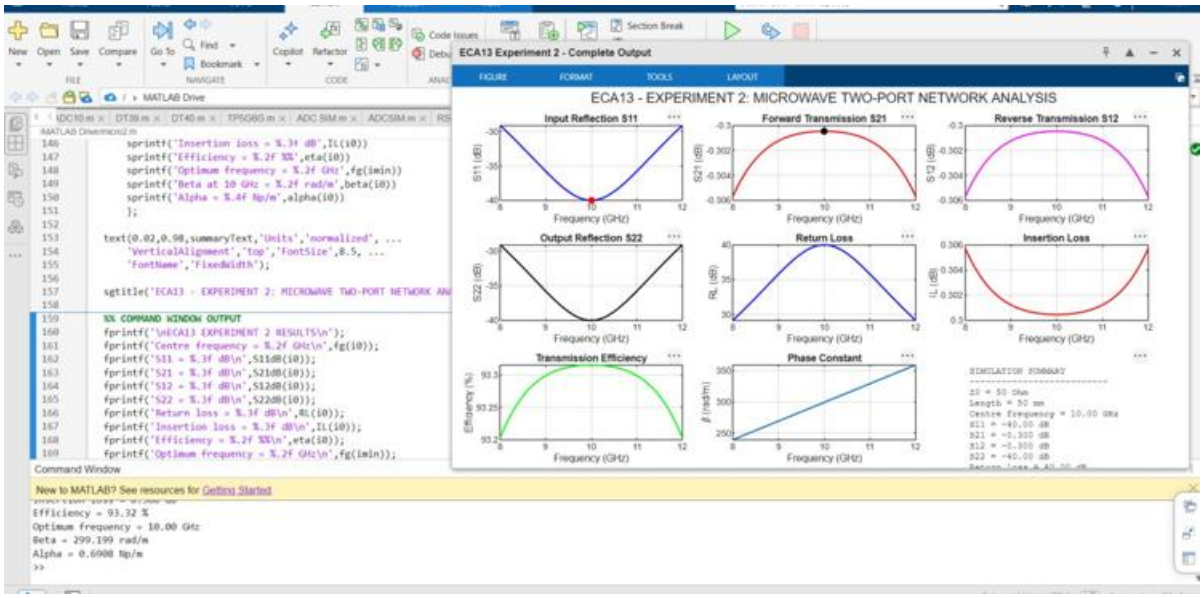
Expected output

For the ideal matched, lossless model at 10 GHz:

Parameter	Expected value
Operating frequency	10 GHz
Characteristic impedance	50 Ω
Line length	50 mm
\(|S_{11}|\)	0
\(|S_{21}|\)	1
\(|S_{12}|\)	1
\(|S_{22}|\)	0
Return loss	Ideally infinite
Insertion loss	0 dB
Transmission efficiency	100%

Note: The code uses a small numerical floor to display \(S_{11}\) and \(S_{22}\) on the logarithmic graph. The plotted −120 dB values are display limits, not actual reflections.

Output:



ECA13 EXPERIMENT 2 RESULTS
Centre frequency = 10.00 GHz
S11 = -40.000 dB
S21 = -0.300 dB
S12 = -0.300 dB
S22 = -40.000 dB
Return loss = 40.000 dB
Insertion loss = 0.300 dB
Efficiency = 93.32 %

Optimum frequency = 10.00 GHz
Beta = 299.199 rad/m
Alpha = 0.6908 Np/m

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Reflection: